

Lab: Building Your Comprehensive Test Plan

Objective

Create a comprehensive test plan for your invention project that prioritizes hypotheses by expected free energy, specifies test protocols, allocates resources, and integrates regulatory requirements. By the end of this lab, you will have a structured document that guides your entire testing phase.

Prerequisites

- Familiarity with expected free energy prioritization, test protocol design, resource allocation, and regulatory testing from this module's lecture
- Your iteration roadmap from Module 05, learning system from Module 06, and documentation framework from Module 07
- A clear understanding of your invention's remaining uncertainties

Materials

- All project documentation from prior modules
- Notebook or digital document for the test plan
- Calculator for resource estimation

Part 1: Test Inventory (10 minutes)

Goal: Create a complete list of everything that needs testing before your invention is ready for its next milestone (whether that is a working prototype, a market launch, or a patent filing).

Brainstorm all testable hypotheses. Do not prioritize yet — just list them all.

#	Hypothesis to Test	Category (Technical / User / Market / Safety / Regulatory)	Currently tested?
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Are there any regulatory or certification requirements for your invention? If so, list them:

Write your response here

Part 2: Priority Ranking by Expected Free Energy (15 minutes)

Goal: Rank your test hypotheses by expected free energy reduction — the combination of uncertainty resolved and consequence addressed.

For each hypothesis from Part 1, rate its uncertainty (how much you do not know) and consequence (how much it matters):

#	Hypothesis	Uncertainty (1-5)	Consequence (1-5)	Estimated Cost	EFE Score (Unc x Cons / Cost)	Priority Rank
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Does your ranking match your intuition? If not, what does the discrepancy reveal about your priorities?

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Part 3: Write a Test Protocol (15 minutes)

Goal: Write a detailed test protocol for your highest-priority hypothesis.

Test Protocol #1

Field	Specification
Hypothesis	

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| | Prediction (specific, falsifiable) |

Write your response here

|| Procedure (step-by-step) |

Write your response here

|| Measurements (what, with what instrument, at what precision) |

Write your response here

|| Sample size (how many tests/users) |

Write your response here

|| Control conditions (what is held constant) |

Write your response here

|| Success criteria (what result confirms the hypothesis) |

Write your response here

|| Failure criteria (what result disconfirms it) |

Write your response here

|| Ambiguity criteria (what result is inconclusive, requiring more testing) |

Write your response here

|| Analysis plan (how will data be interpreted) |

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Review: Is this protocol specific enough that someone else could execute it and get comparable results?

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Part 4: Resource Allocation (10 minutes)

Goal: Map your available resources to your testing needs.

Available resources:

Resource	Available Amount	Notes
Time		

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| | Budget |

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|| Materials/Equipment |

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|| Testers/Users |

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|| Expertise |

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Resource allocation to top-priority tests:

Test (by priority)	Time Needed	Cost Needed	Materials Needed	Can run in parallel with?	Dependencies (must complete before?)

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Critical path: Which test sequence determines the minimum total testing time?

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Testing debt check: Are there any high-consequence tests you are tempted to defer? What is the risk of deferring them?

Write your response here

Part 5: Assemble the Test Plan (10 minutes)

Goal: Compile everything into a one-page test plan summary.

Test Plan for:

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Objective: What milestone does this test plan support?

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Priority-ordered test sequence: 1.

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Total estimated time:

Write your response here

Total estimated cost:

Write your response here

Decision gates: - After Test #___: Proceed / Iterate / Pivot decision based on:

Write your response here

• After Test #___: Proceed / Iterate / Pivot decision based on:

Write your response here

Regulatory requirements integration:

Write your response here

Risk register (what could go wrong with the testing process itself):

Risk	Likelihood	Impact	Mitigation

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Discussion and Debrief

1. **Priority surprises:** Did the expected free energy ranking produce a different testing order than your intuition? What does this suggest about how you naturally prioritize?
2. **Protocol discipline:** Was it difficult to specify success and failure criteria in advance? What is the temptation you feel to leave these vague?
3. **Resource reality:** After mapping resources to testing needs, do you have enough to test everything? What trade-offs are you forced to make?
4. **Testing debt:** Are you carrying any testing debt — important tests you have deferred because they are inconvenient or expensive? What is the risk?
5. **Adaptability:** How will your test plan change as early tests resolve uncertainties? Is the plan structured enough to provide guidance but flexible enough to adapt?

Write your response here